

Training Configuration and Summary:

Sequencing:

- timesteps: 30
- batch_size: 256

Model Hyperparameters:

- min_units: 16
- max_units: 64
- units_step: 16
- dropout_min: 0.1
- dropout_max: 0.5
- dropout_step: 0.1
- activation_functions: ['tanh', 'relu', 'elu']
- optimizers: ['adam', 'rmsprop', 'Nadam']

Tuning:

- max_epochs: 30
- hyperband_iterations: 3

Directories:

- directory: my_dir
- project_name: lstm_hyperparameter_tuning

LSTM Training Report

Report Generated: 2025-04-08_14-16-09

Training Start: 2025-04-08 14:10:45

Training End: 2025-04-08 14:12:23

Training Duration: 97.59 seconds

Best Hyperparameters:

Number of Layers: 1

Activation: tanh

Optimizer: Nadam

Dropout Rate: 0.30

Units per Layer:

Layer 1: 128 units

Performance Metrics:

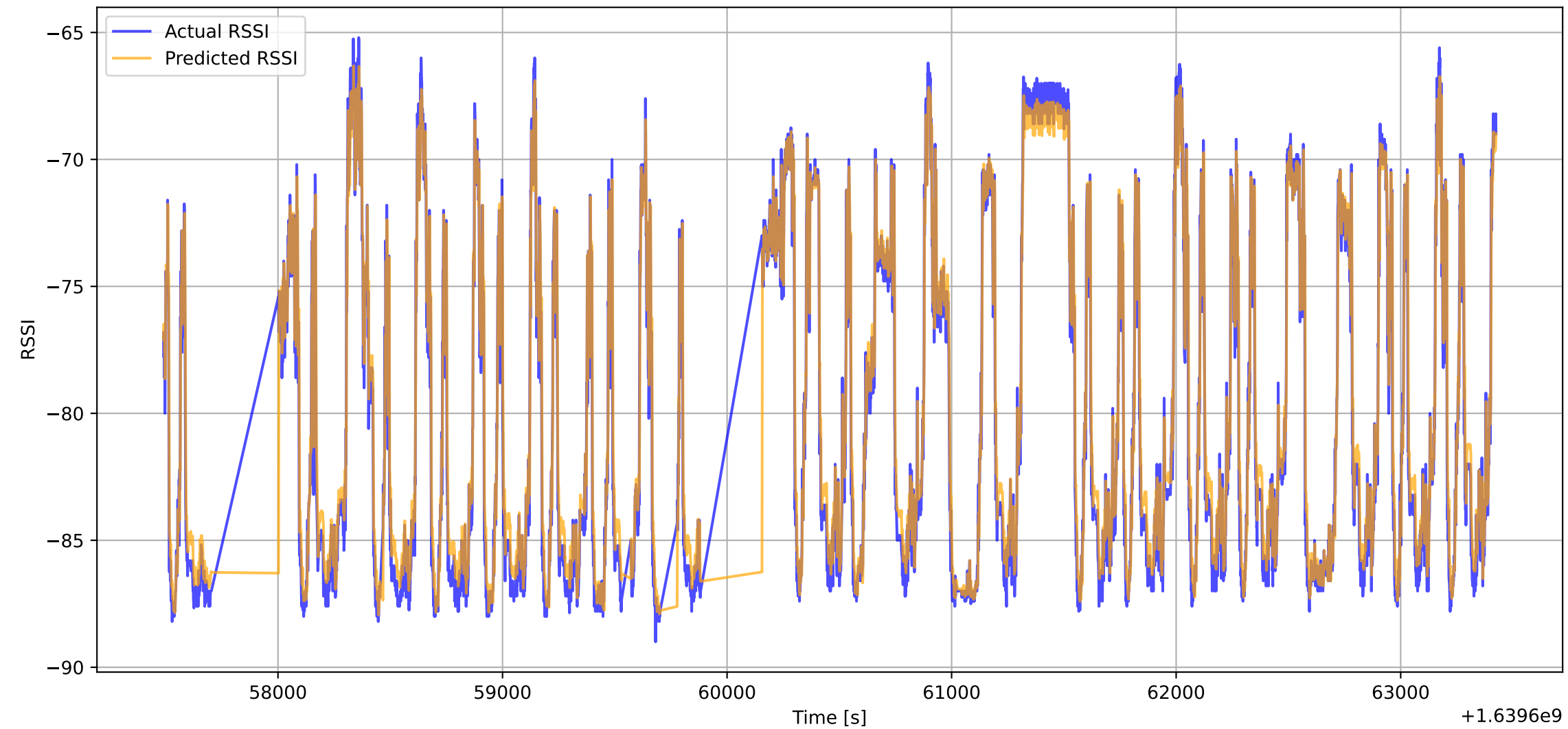
MAE: 1.0100

MSE: 1.9420

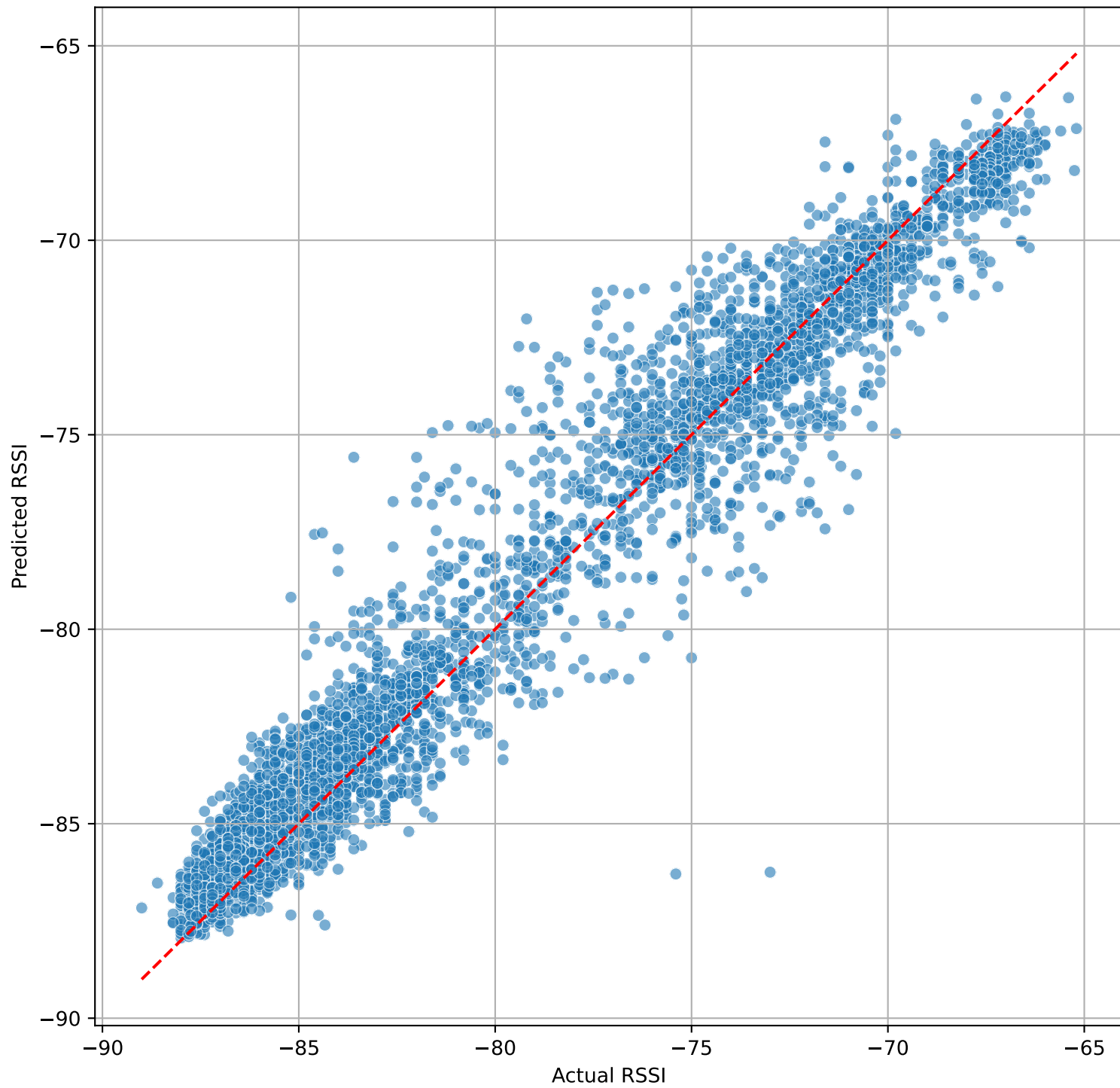
RMSE: 1.3935

R2 Score: 0.9559

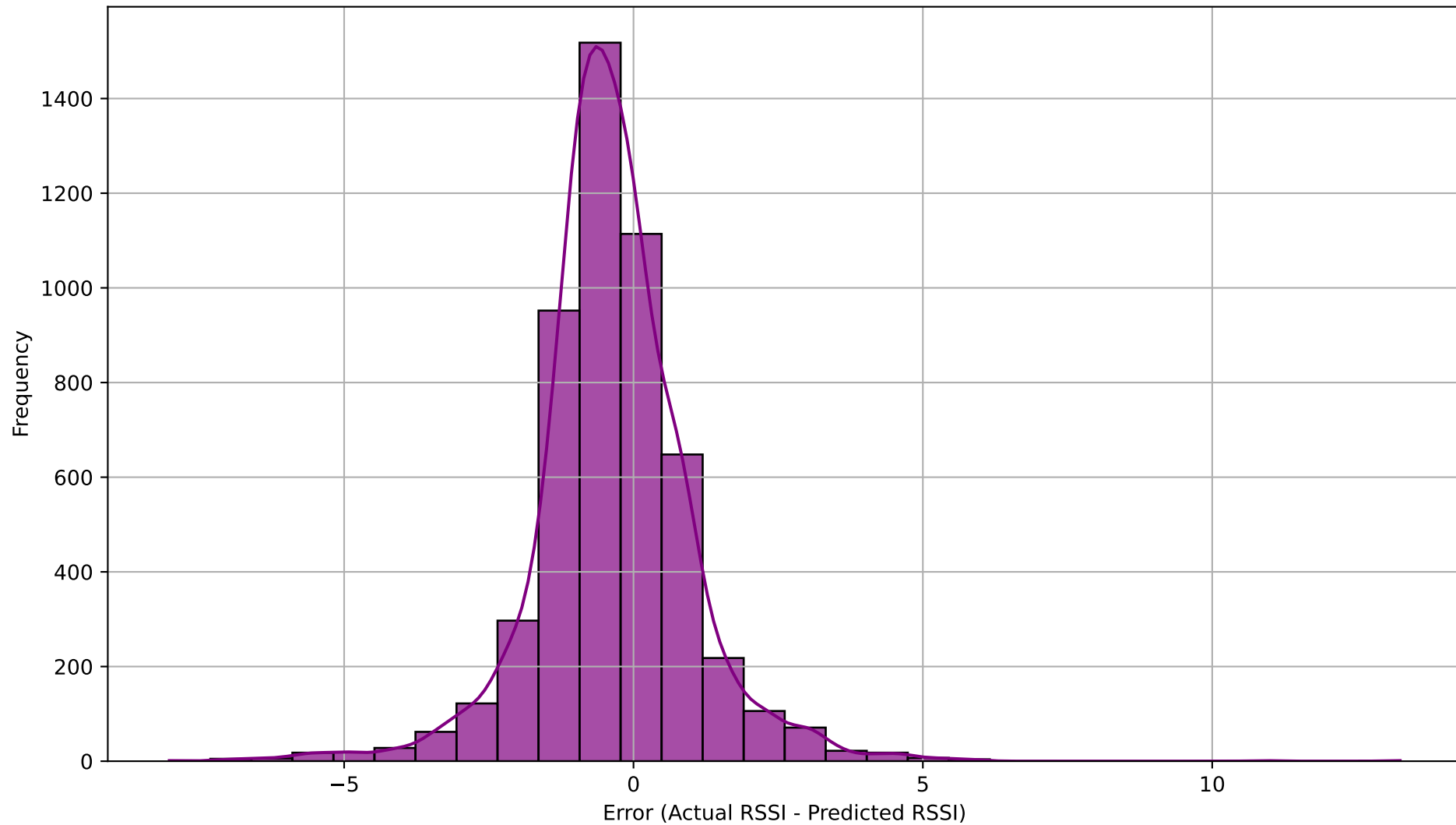
Actual vs Predicted RSSI Over Time



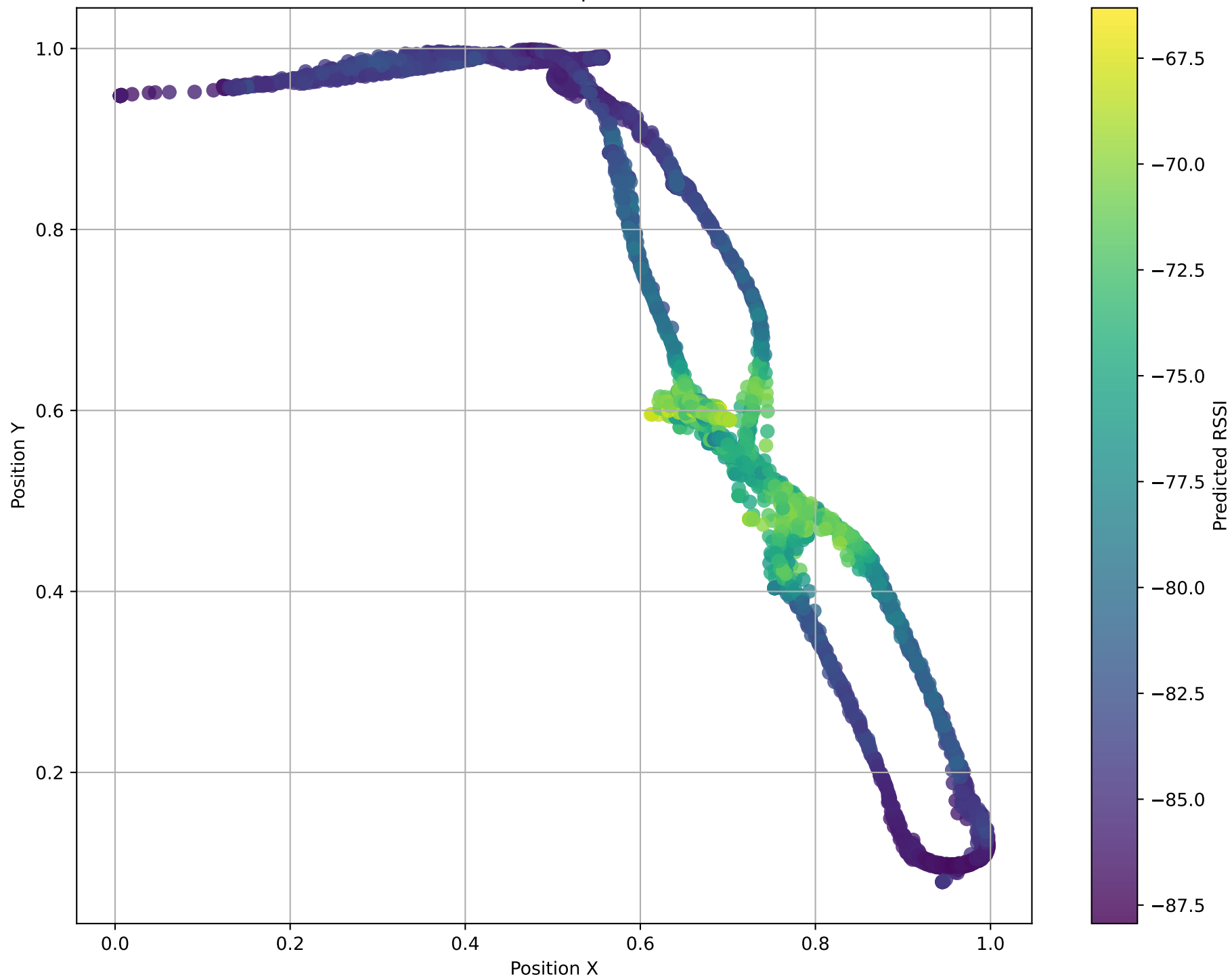
Actual vs Predicted RSSI



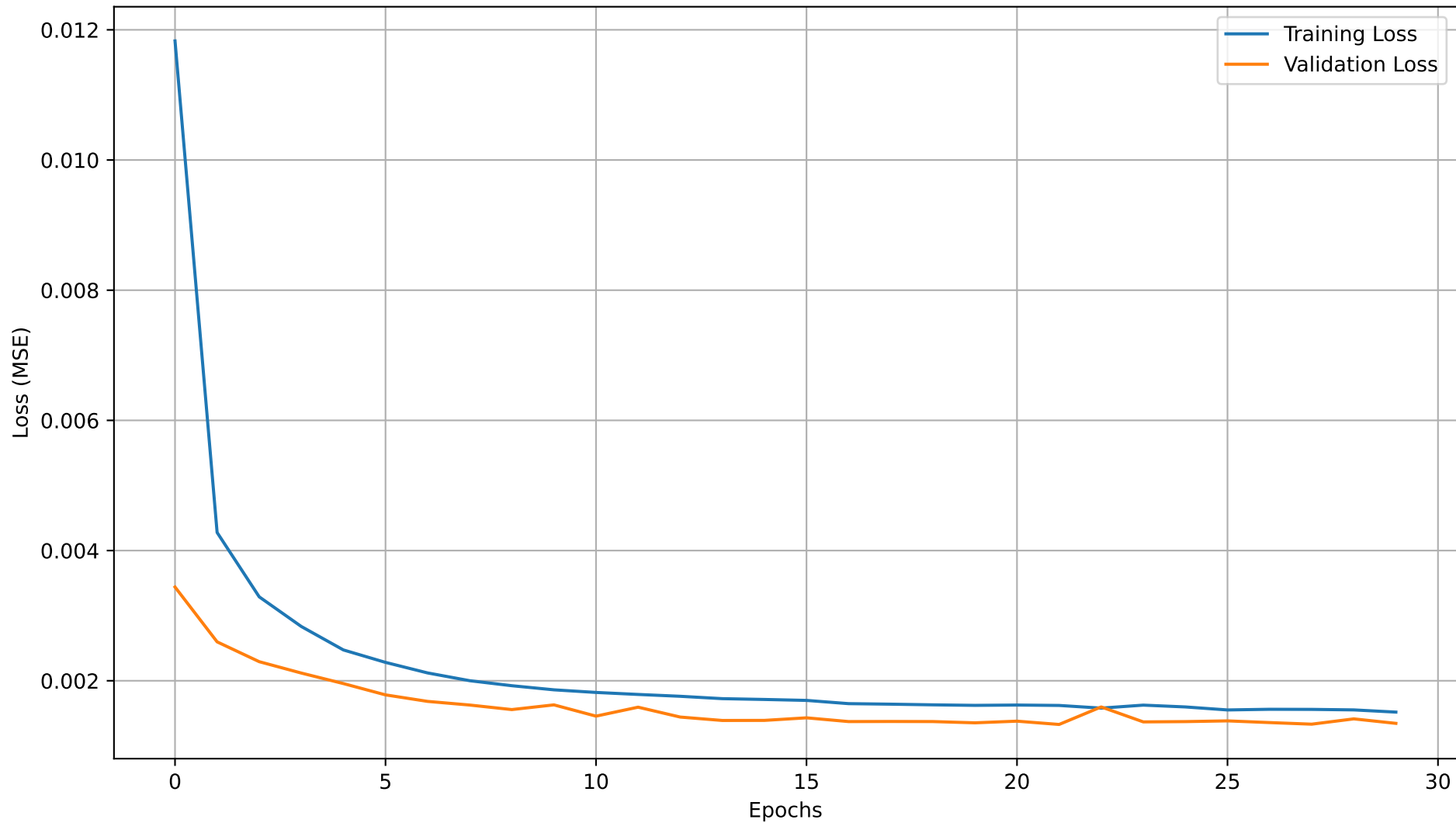
Distribution of Prediction Errors



Predicted RSSI in Spatial Context



Training and Validation Loss Over Epochs



Training and Validation MAE Over Epochs

